Proposed Solution

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
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Project overview

Trip Genius AI (also referred to as VoyagePlan Travel Planner during development) is a web-based travel planning application that turns user preferences into a structured trip itinerary. Users enter a destination, travel dates, interests, budget and pace; the application sends the request to a FastAPI backend that uses Groq AI to produce the itinerary. Signed-in users can save trips, revisit them and edit their preferences.

Objective

Reduce the manual effort of organising a multi-day trip by providing a clear itinerary-planning workflow with personalisation inputs and persistent trip management.

Problem statement

Travellers often compare information from multiple websites and manually organise activities into a practical day-by-day plan. Saved notes become fragmented and changing preferences requires repetitive work.

Proposed solution and key features

Feature	Description
Authentication	Supabase-based sign-up and sign-in.
Trip preference form	Destination, dates, interests, budget and pace, with validation.
Itinerary generation	Frontend submits preferences to POST /api/itineraries/generate; FastAPI calls Groq and returns the itinerary.
Itinerary display	Shows the trip preferences and day-wise content.
Saved trips (My Trips)	Stores trips per user in Supabase and lists them for later access.
Edit preferences	Reopens a saved trip with its prior values restored.
Deployment	Vercel frontend, Render backend, Supabase services.

Scope and limitations

The application is an itinerary planning assistant, not a booking, payment or live-inventory platform. Activity availability, opening hours, travel times and prices should be independently checked because AI output can be incomplete or outdated.

Expected outcome

A deployed application with an end-to-end workflow for creating, saving, revisiting and editing travel plans. The core workflow has passed production smoke tests; user studies and quantitative performance testing remain future work.